Environmental Selection and the Transitional Role of Antiparticles in Dynamic Equilibrium

Version 2 – Compliance Revision

Geoffrey E. Thwaites
Surfers Paradise, Queensland, Australia

Core Claim

Environmental conditions determine whether newly created particle–antiparticle systems follow a matter-retention pathway or an energy-return pathway.

Definitions

Environmental Selection, Matter Retention, Energy Return, Dynamic Equilibrium, and the Law of Inevitability are defined as operational concepts governing outcomes.

Mechanism

Energy → Particle–Antiparticle Production → Environmental Selection → Matter Retention + Energy Return → Dynamic Equilibrium.

Observational Basis

Particle–antiparticle pairs exist, stable matter persists, energy-return pathways exist, and environmental conditions influence outcomes.

Testable Hypothesis

H1: Survival probabilities depend on environment. H0: They do not.

Test Path

Compare retention and return outcomes while varying environmental conditions.

Falsification

The hypothesis fails if retention and return probabilities remain unchanged under controlled environmental variation.

Predictions

Retention-return ratios vary with environment; matter-rich environments favour retention pathways.

Conclusion

The environment is the selector. The resulting state is the inevitable consequence of the conditions present.

Addendum A – Proposed Experimental Evaluation Pathway

Suitable Facilities

CERN Antimatter Factory facilities including ALPHA, AEgIS, BASE, ELENA and the Antiproton Decelerator.

Experimental Concept

Generate controlled antiparticle populations, vary environmental conditions, and measure survival, retention, and annihilation rates.

Predicted Outcome

Environmental change should alter retention-return ratios.

Status

Conceptual test design; proposed as a pathway to empirical evaluation.